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Emotional and social well-being of gifted adolescents

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Abstract

This study examined the social functioning of people with characteristics of giftedness compared to non-gifted year-level peers within higher-level secondary education. Mental health problems, emotional intelligence, feelings of competence, feelings of (social) (in)adequacy and ego development were surveyed among four groups of adolescents (N range from 252 to 545 secondary school students) of varying intellectual capacity, extending from the high-average and superior ranges into the highly gifted range: high-average to superior intelligent, near gifted, gifted and highly gifted. The emotional and social well-being of adolescents in the gifted and highly gifted range did not differ from the other adolescents in the upper segment of the ability spectrum. The findings support other studies and disprove the assumption of vulnerability and maladjustment in adolescents in the gifted and highly gifted range, which might contribute to more awareness about persistent stereotypes that might hinder the development of adolescents with characteristics of (highly) giftedness.

KEYWORDS

competence, ego development, emotional intelligence, feelings of (in)adequacy, intellectually gifted, mental health

Key points

- This study was composed of a sample of adolescents recruited from a single year level across multiple schools with varying intellectual capacity, extending from the high-average and superior ranges into the highly gifted range without self- or pre-selection of gifted individuals.
- This study demonstrates that the emotional and social well-being of highly gifted, gifted and near gifted adolescents was equally likely in terms of mental health, interpersonal intelligence, adaptability, positive impression, social acceptance, close friendship, feelings of (social) inadequacy and ego-development and also when compared with their year-level peers.
- Prior research exhibited a tendency toward the harmony hypothesis, a pattern similarly observed in the present study. The added insight from this study is that persistent stereotypes of vulnerability or maladjustment are additionally refuted within the gifted population, and within the grade cohorts in which the gifted students are located.
- Practical implications: The risk of overgeneralizing the vulnerability of (highly) gifted people by many adults, parents, newspaper articles and (educational) professionals in society underlines the importance of (psycho)education,

of individualized, strength-based perspectives and of paying attention to the broader ecological (mis)conditions that determine emotional and social outcomes for (highly) gifted people.

INTRODUCTION

From its inception, research on giftedness has explored and discussed diverse perspectives on the difference between the well-being of the gifted and non-gifted (Rinn, 2024). Even though more insights are still needed to gain a more comprehensive view on giftedness and specific characteristics of gifted people, there are some brain studies pointing at differences between gifted and non-gifted people, suggesting that brain development in gifted individuals may differ from that of their non-gifted peers. High intelligence appears to be associated with more efficient neural processing, particularly in the left hemisphere, denser fibre networks and a brain organization that promotes effective inter-regional communication (Barbey, 2018; Solé-Casals et al., 2019; Tetreault & Zakreski, 2020). Gifted children often show earlier activation of frontal, parietal, temporal and occipital regions, while certain areas of the prefrontal cortex may mature later, particularly around the onset of puberty (Jung & Haier, 2007; Shaw et al., 2006; Suprano et al., 2019). In addition, early neurophysiological and psychomotor development has been observed in intellectually gifted children and adolescents (Vaivre-Douret, 2004, 2011; Vaivre Douret et al., 2020). Question is if, apart from these neurofysiological differences, gifted people also differ from non-gifted people in emotional and social development. In scholarly literature, two longstanding paradigms have addressed how people perceive the characteristics of (highly) gifted individuals (Hollingworth, 1942; Terman, 1925–1959). Firstly, the harmony hypothesis, which emphasizes elevated capacities, a more balanced personality profile and consistently high performance outcomes. Secondly, the disharmony hypothesis, which recognizes similar levels of intellectual potential and accomplishment but concurrently associates giftedness with an increased prevalence of social and emotional maladjustments (Barzut et al., 2020; Baudson, 2016).

Several empirical research supports the harmony hypothesis indicating that gifted people benefit from their high intelligence, which means that their scores deviate marginally or are higher compared to non-gifted people on a variety of emotional and psycho-social variables. The high intelligence of gifted individuals enables them to experience more succes, use more adaptive coping mechanisms and experience higher social competence feelings (Bergold et al., 2015, 2020; Jokela, 2022). Several studies show that gifted individuals function quite well socially (e.g. Falck, 2020; Papadopoulos, 2020). Many present-day giftedness experts lean toward the view that (intellectually) gifted individuals as a group are well-adjusted, socially accepted and generally do not experience more social or emotional problems than their same-age non-gifted peers (Bergold et al., 2020; Francis et al., 2016; Freeman, 2013; Gross, 2002; Jones, 2013; Martin et al., 2010; Reis & Renzulli, 2004). Neihart's review placed the disharmony hypothesis opposite the harmony hypothesis and determined that empirical backing for the harmony perspective was markedly more robust (Neihart, 1999).

Nevertheless, this does not mean gifted children and adolescents are immune to social and emotional problems (Kennedy & Farley, 2018). For example, a mismatch can arise when gifted students are expected to adapt to environments designed to fit their non-intellectually gifted peers of the same age (Robinson & Reis, 2016).

There is less support for the disharmony hypothesis. Some empirical evidence has been found to support the assumption that intellectually gifted individuals do not function well socially (Peterson, 2009). A more recent longitudinal research study on intellectually gifted adults indicates that an increase in cognitive intelligence does not parallel with better mental health (Czerwiński et al., 2024). Highly gifted individuals may experience unique challenges that may outweigh the benefits of their high intelligence. However, it is important to emphasize that the nonlinearity of this relationship does not mean that they have poorer mental health than the general population. Rather, the nonlinearity suggests a more complex interplay of factors beyond intelligence alone (Czerwiński et al., 2024).

Despite most research pointing in the direction of the harmony hypothesis, the disharmony hypothesis seems to be widespread in many parents, in newspaper articles (Bergold et al., 2021), in adults (especially male, single parents, unemployed, to a certain extent with high income and a negative attitude against the gifted) (Baudson, 2016) as well as among educational professionals (Matheis et al., 2019; Sanchez et al., 2022; Weyns et al., 2021). Multiple studies have found that respondents—including school counsellors in pre-service (in training), (pre-service, in-service) teachers and a broad general population of adults—primarily held negative stereotypes about gifted individuals, ranging from young children to adults (Baudson, 2016; Baudson & Preckel, 2013, 2016; Matheis et al., 2017, 2019; O'Conner, 2005). Research of Barzut et al. (2020) found that preschool teachers are generally not particularly positive about the social adjustment of gifted children—their attitudes ranged from neutral to slightly negative. These findings partially support the disharmony hypothesis.

Several reasons have been given as to why these prejudices persist. Robinson (2002) suggested that these beliefs may be primarily attributable to a lack of familiarity with intellectually gifted adolescents. For example, more experienced teachers have been shown to make fewer assumptions that are not in line with research findings about gifted students on the disharmony hypothesis than less experienced teachers (Carman, 2011).

Bergold et al. (2021) referred to the persistent stereotyped representation of gifted individuals in newspaper articles as a factor contributing to these negative perceptions. Also, the contrast of an asynchronous development (precocious cognitive development and age-appropriate emotional and social development) (Papandreou et al., 2023) may lead observers to misjudge the age-appropriate abilities as deficient (Baudson & Preckel, 2016; Bergold et al., 2021).

The ongoing nature of these prejudices can have significant consequences. These stereotypes often simplify the complex realities of gifted individuals into generalized assumptions (Baudson & Preckel, 2016). Also, stereotypes can influence teachers' expectations and subsequently shape students' behaviour (Matheis et al., 2019). Expectations of important others (as teachers) stemming from stereotypes can lead to mistaken assumptions, adversely affecting gifted individuals' social experiences and imposing psychosocial risks and can have a negative impact on the development of gifted youth (Bergold et al., 2021). Furthermore, such challenges may result in defence mechanisms, such as masking or rationalization and hindered development (Bergold et al., 2021; Frumau-van Pinxten et al., 2020; Gross, 2002; Rimm, 2002; Ruf, 2009; Tolan, 2018b).

The aim of this study was to obtain deeper insight into potential differences among secondary school students, situated at the upper end of the cognitive ability distribution and attending the highest educational tracks, including distinctions *within the gifted population*, most notably between gifted and highly gifted adolescents, in line with the harmony or disharmony hypotheses. More specifically, we wanted to gain insight into self-perceived differences of four groups of secondary school students (intellectually highly gifted, gifted adolescents, near gifted and high-average to superior intelligent adolescents) regarding their strengths and difficulties, emotional intelligence, feelings of competence, feelings of social (in)adequacy, social (in)adequacy and ego development.

First, the adolescents' *mental health* encompasses their strengths and challenges. The more overlap between the self and the self-concept, including perceived strengths and challenges, the more psychologically healthy these adolescents are and will become as adults. Martin et al.'s (2010) review on nine studies found that gifted youth show much less anxiety than non-gifted peers and have similar levels of depression and suicidal thoughts. Francis et al.'s (Francis et al., 2016) systematic review found that, overall (two out of the 18 studies reported negative findings and four studies reported a mix of positive and negative findings about the link between intellectual giftedness and mental health problems) intellectually gifted children tended to be better socially adjusted and experienced fewer emotional or behavioural problems. Eren et al. (2018) found comparable scores between gifted and non-gifted adolescents regarding mental health problems (SDQ), with one exception: the gifted claimed to be slightly more inattentive/hyperactive.

Based on those findings we expect no difference between the groups.

Second, *emotional intelligence* is a relevant concept to investigate in gifted individuals, because it is the ability to identify and regulate one's emotions and understand the emotions of others and, therefore, the critical ability to relate to self and others (Perrotta & Basiletti, 2023). Previous studies on emotional intelligence in the gifted found it equivalent or higher scores in the gifted compared to non-gifted students (Abdulla Alabbasi et al., 2020; Ogurlu, 2021; Zeidner et al., 2005). Based on these findings we expected that the highly gifted and gifted adolescents show higher emotional intelligence, especially higher levels of *adaptability*, than the other adolescents. This expectation was based on the assumption that the gifted must compensate for the almost inevitable mismatch with their environment (Robinson, 2002).

Third, feelings of social acceptance and close friendship as *feelings of social competence* seem relevant to social and emotional well-being. Research on social acceptance among gifted individuals shows mixed results. Košir et al. (2015) found no significant differences in social acceptance and self-concept between gifted and non-gifted adolescents. In contrast, Kroesbergen et al. (2016) reported lower social acceptance for gifted first and second graders compared to their non-gifted peers in the Netherlands. Additionally, highly gifted students appear to face more social acceptance challenges than moderately gifted ones (Rimm, 2002). Shore et al. (2019) emphasized the importance of having at least one close friend for gifted individuals, Shore et al. (2025) cited Masden et al. (2015) who cited previous claims (e.g., Gross, 2003; Silverman, 1993) that exceptionally gifted children and youth had fewer friends, unless they could connect with others of similar interests and abilities. They also summarized several studies finding gifted friendships to be somewhat different and specifically that these gifted friendships focus on avoiding competition or competing for fun (Shore et al., 2019).

Based on the findings on adolescents of Košir et al., Rimm and Shore, we expect no difference between the groups on social acceptance, except for the highly gifted. For the highly gifted, we expect reduced social acceptance and close friendship.

Fourth, *feelings of (social) (in)adequacy* can affect adolescents' social and emotional well-being. Feelings of inadequacy encompass fears, a depressed mood, physical complaints with unclear causes and feelings of inferiority. Feelings of social inadequacy encompass feeling unhappy with and avoiding social contact (Luteijn et al., 2005). Some earlier studies did not find an increased level of anxiety in the face of challenges or possible failure and, thus, *feelings of inadequacy* among the gifted (Peterson, 2009; Webb et al., 2016). However, Eren et al. (2018) found giftedness slightly but not significantly affected their overall quality of life and more depressive moods in gifted boys. Based on the findings on adolescents of Peterson and Webb, we expect no difference between the groups on social (in)adequacy.

Finally, it is known that *ego development* is an indicator of social–emotional development. Ego development refers to how an individual perceives the world, experiences social interactions and forms relationships with others (Loevinger, 1976). Bailey (2011) found that gifted adolescents (grades 9–12 in Virginia, USA) were not advanced in their ego development although sometimes, but not consistently, they scored slightly higher than normative levels of ego development. In their meta-analysis, Cohn and Westenberg (2004) found a faster ego development in adolescents with a higher cognitive intelligence, meaning that their rate of ego development is higher. Based on this research, we expect higher levels of ego-development in the gifted.

Few studies have examined a broad range of characteristics of highly gifted and gifted adolescents. The existing studies appear to lack consistency. Moreover, studies investigating the *(dis)harmony hypothesis* often used preselected or self-selected groups of gifted adolescents (Lavrijsen & Verschueren, 2023). Highly gifted adolescents were usually neither examined separately nor compared with gifted, near gifted and high-average to superior intelligent adolescents.

For the above reasons, the present study examined a cohort of students in higher-level secondary education in the Netherlands. As Martin et al. (2010) advised, a large, population-based cohort was examined to be as representative as possible for the group of cognitively most intelligent Dutch adolescents. They argued that ideally, research on assessing the risk of mental health issues among gifted individuals should include a sample of (highly) gifted and non-(highly) gifted youth assessed at the same time, with the same mental health measures, using the same methods, by the same study team. In line with this suggestion, this study used a sample of adolescents with varying intellectual capacity, extending from the high-average and superior ranges into the highly gifted range without pre-selection of gifted individuals.

Problem statement

Long-standing societal stereotypes regarding perceived social and emotional weaknesses have been associated with gifted and highly gifted people. This development is notable given that the field increasingly aligns with the harmony hypothesis, which emphasizes balanced rather than deficient emotional and social functioning. However, very few studies have examined whether differences actually exist within the upper segment of the ability spectrum or within the gifted population, leaving open the question of whether any empirical basis for such persistent stereotypes can be found in the social characteristics of the highly gifted. Establishing research within the upper segment of the ability spectrum and specifically within the gifted population on the presumed vulnerabilities is crucial: if no differences are found, this provides stronger grounds to challenge the validity of these stereotypes. Moreover, it underscores the importance of shifting attention away from presumed individual weaknesses of gifted and highly gifted individuals and toward the broader ecological conditions that shape emotional and social outcomes.

Research question

This study addresses the following research question:

In what way do the four groups (intellectually highly gifted, gifted adolescents, near gifted and high-average to superior intelligent adolescents) differ in strengths and difficulties, emotional intelligence, feelings of competence, feelings of social (in)adequacy, social (in)adequacy and ego development?

This study is situated within a dual theoretical framework consisting of (1) the harmony hypothesis within the gifted population and (2) perspectives on the prejudices and stereotyping of gifted and highly gifted adolescents. These perspectives provide the conceptual foundation for the present study and position it within the current state of the art. Based on this framework, no differences concerning strengths and difficulties, feelings of competence, feelings of social (in)adequacy and social (in) adequacy were expected between intellectually highly gifted, gifted and non-gifted adolescents, with one exception. We expect the highly gifted to experience less social acceptance and less close friendship. Also, higher scores in emotional intelligence (especially adaptability) and ego development for the gifted and highly gifted were expected.

METHOD

Design

This study is part of a more extensive longitudinal study, using self-report questionnaires administered to adolescents at two time points with an interval of 2 years. Ethical approval for the present study was granted by the ethics committee of Fontys Ethics Committee.

Participants

In this study, a large, non-selective sample of adolescents participated: 252–545 students in Dutch grade 8, equalling the second year in secondary education in the Netherlands (age 11–15 years, average age: 13.58). All eighth graders from six regular secondary schools and their parents were informed about the research aims by an information letter and were invited to participate. They were not selected from gifted programs or based on teacher recommendations because teacher nominations can be biased (Golle et al., 2023). All participants were approached by their schools. At the end of the study, all parents of the participants received a summary report of the findings.

The study involved six schools in a small region in the south of the Netherlands, offering ten levels of education. Dutch secondary education has three main levels: VMBO (vocational training, 4 years), HAVO (prepares students for higher professional education (HBO), 5 years) and VWO (academic preparation, 6 years). VWO includes athenaeum and gymnasium, with the latter requiring Latin, Greek and ancient culture studies. Most students begin secondary school at age 12, often after an orientation year. Some schools provide bilingual education in Dutch and English. All schools offered an academic track of secondary education, and three participating schools operated exclusively as VWO gymnasia. The participants in this study were enrolled in higher-level secondary education, represented the highest-performing 14% of the population, rather than the population mean and are presumed to span the cognitive spectrum from the high-average and superior ranges into the highly gifted range.

Participants were divided into groups according to their cognitive intelligence level (further explained in the next paragraph). Because of item nonresponse, the sample size was not identical across the instruments administered. For (group 1) the intellectually highly gifted, SDQ, EQi-YV and CBSA scores were obtained for $N=24$ participants, ZALC scores for $N=19$ and NPVJ-scores $N=11$. For (group 2) the intellectually gifted, SDQ, EQi-YV and CBSA scores were obtained for $N=48$ participants, ZALC scores for $N=35$ and NPVJ-scores $N=16$. For (group 3) the near gifted, SDQ scores were obtained for $N=111$, CBSA and EQi-YV for $N=107$ participants, ZALC scores for $N=67$ and NPVJ-scores $N=44$. For (group 4) the high-average to superior intelligent, SDQ scores were obtained for $N=404$, CBSA and EQi-YV for $N=369$ participants, ZALC scores for $N=275$ and NPVJ-scores $N=181$. The *Raven APM* (Raven, 2008; Raven & Court, 1998) was used to identify the cognitive intelligence level. The *Raven APM* is suitable for individuals aged 12 years and up (secondary school students, higher education students and adults) with high-average to high cognitive abilities and thus well-suited for children with high cognitive abilities, due to the 'out-of-level-testing' idea (Warne, 2012). It is the most difficult version, measuring a high level of abstract reasoning. The *Raven APM* calls upon general reasoning skills, which are important for verbal (linguistic) and nonverbal (nonlinguistic) cognitive functioning and are closely related to the *g*-factor. Unlike other tests of cognitive capacity, the *APM* (i.e. advanced) version of the Raven test does not have a noticeable ceiling effect for high intelligence. If the *Raven SPM* test had been used, the highly gifted, gifted and probably the near-gifted adolescents were expected to reach the test ceiling, also known as the 'ceiling effect' (Assouline & Lupkowski-Shoplik, 2012).

In this study, giftedness has been operationalized as general cognitive ability (Spearman's g-factor (1904)), the most profound aspect of intelligence relevant to solving many different cognitive tasks. This *g* is a robust factor, meaning it is not strongly influenced by cultural differences and seemingly an universally present human feature (Warne & Burningham, 2019). The intellectually highly gifted adolescents were identified using the *Raven APM* test (Raven, 2008; Raven & Court, 1998) administered in grade 8 in this research.

Cut-off points were chosen based on two presumptions to divide students into groups: (1) the levels in high-level cognitive intelligence or gifted levels of Ruf's research (2021); and (2) the cut-off point that is widely used in the literature on giftedness. These two presumptions were considered to align coherently in this study. Regarding the first presumption, the intellectually highly gifted adolescents in group 1 were identified using a cut-off point of up and above about percentile 98, equaling Ruf's gifted levels two to five, which is equivalent to a score on the RAVEN of 30 or higher. Group 2 comprised the intellectually gifted adolescents in percentiles 90–98, the approximate average of Ruf's gifted level one (Ruf, 2021) (RAVEN APM score of 29). Group 3 consisted of the near gifted adolescents in percentiles 75–90 (RAVEN APM score of 28) and for group 4, the high-average to superior intelligent adolescents, a cut-off point of lower than percentile 88, equivalent to a score of <28 on the RAVEN APM, was applied. Although Ruf's framework is not an empirically validated classification system, we consider it useful for examining characteristic differences within subgroups of intellectually gifted students. Regarding the second presumption, percentile 95 is often used in research and identification of giftedness (e.g. Godor & Szymanski, 2017), equalling an IQ of 124.7 with a 1.65 SD. According to the German norming data of the RAVEN APM, a score of 29 corresponds to the 95th percentile. Whereas a raw score of 30 corresponds to the 98th percentile. A raw score of 28 was considered near gifted and raw scores below 28 (e.g. 27 and down to approximately 24, which corresponds to roughly the 74th percentile) were classified as *high average to superior intelligence*. In a standard Gaussian distribution, scores from approximately the 75th percentile onward fall within the above-average range.

This classification provides a norm-referenced and transparent foundation for distinguishing between different levels of intellectual high ability within our sample.

Instruments

All instruments are presented in Table 1, including scale descriptions, example items and internal consistency.

Secondly, the self-report version of the *Strengths and Difficulties Questionnaire* (Richter et al., 2011) was applied. This questionnaire investigates mental health problems. Each item can be answered with 'not true' (0), 'somewhat true' (1) or 'certainly true' (2). Several studies have evaluated the internal consistency of the self-report version as satisfactory (Richter et al., 2011). The results of this study raised some questions about the interpretation of some subscales but not the total score. This was reason to use only the total difficulty score in this study.

Thirdly, the participants completed the *Emotional Quotient Inventory: Youth Version* (EQi-YV) for children and adolescents between the ages of 7 and 18 (Bar-On & Parker, 2000). For the present study, three of the six subscales concerned with social intelligence were used: the Interpersonal scale, the Adaptability scale and the Positive impression scale. Items were responded to on a four-point Likert scale ranging from 1 = 'very seldom or not true for me' to 4 = 'very often true or true for me.' The *EQ-i:YV* has a replicable factor structure (Bar-On & Parker, 2000). This study also used its measurement of positive impression to examine the tendency to show more social desirability in self-reports.

Fourthly, the Dutch version of the *SPPA* (*Self-Perception Profile for Adolescents*; Harter, 1988) was completed, the *Competentie Belevingsschaal voor Adolescenten* (*CBSA*). This questionnaire identifies social and emotional problems among adolescents aged 12–18 using six specific scales. The *Social Acceptance* and *Close Friendship* scales were used for the present study. Items consist of two statements, each offering a choice between statements A or B, in which the answer is 'a bit true for me' or 'completely true for me.' As the name indicates, the items composing the *Social Acceptance* scale refer to the experience of feeling socially accepted by others. The scale of *Close Friendship* refers to the extent to which adolescents consider themselves capable of maintaining close friendships or the extent to which they can share secrets with friends. The internal consistency of the two scales has been found to be acceptable (Harter, 1988).

Fifthly, the *Nederlandse Persoonlijkheidsvragenlijst* (*NPV-J*) for 9- to 16-year-olds [*Dutch Personality Questionnaire*] of Luteijn et al. (2005) was completed.

TABLE 1 All instruments. Overview instruments, subscales, number of items, alpha, scale description and examples of items.

Instruments	Subscales, number of items, alpha	Scale description	Example of items
Raven APM[a] (Raven & Court, 1998)	36 items $\alpha = 0.82$	Non-verbal intelligence test measuring logical reasoning ability (or rule-sensing reasoning)	
SDQ (Richter et al., 2011)	25 items Subscales (all 5 items): Emotional symptoms; Conduct problems; Hyperactivity-inattention and Peer problems. Total scale (range 0–40). $\alpha = 0.70$ to 0.81	Internationally used screening instrument for mental health problems in children and adolescents	'Often unhappy, downhearted… (*I am often unhappy…*)' 'Considerate of other people's feelings (*I try to be nice to other people*)'
EQi-YV (Bar-On & Parker, 2000)	Subscales: the *Interpersonal scale* (12 items); the *Adaptability scale* (10 items); the *Positive impression scale* (6 items). $\alpha = 0.84$ to 0.89	Emotional intelligence, in particular interpersonal functioning, adaptability, positive impression	'I am good at understanding how other people feel'
CBSA (Harter, 1988)	Subscales (all 5 items): *Social Acceptance and Close Friendship.* $\alpha = 0.76$ and 0.73	Social and emotional problems among adolescents	'Some young people are not easily liked' 'Some young people can keep a good friendship for a long time'
NPVJ (Luteijn et al., 2005) *ZALC* (Westenberg et al., 2000)	Subscales: the *Inadequacy scale* (28 items) and the *Social Inadequacy scale* (13 items). $\alpha = 0.89$ and 0.73 32 items $\alpha = 0.95$	The expression of neuroticism and of social behaviour Ego development	'I often think I am not worth anything' 'There are only a few people who understand me.' 'I don't like talking to strangers' 'I have a hard time making new friends.' 'My father…' 'Rules are…'

Abbreviations: CBSA, Competentie Belevingsschaal voor Adolescenten [Competence Experience Scale for Adolescents]; EQI-YV, Emotional Quotient Inventory: Youth Version; NPVJ, Nederlandse Persoonlijkheidsvragenlijst [Dutch Personality Questionnaire]; ZALC, Zinnenaanvullijst Curium [Sentence Completion Test of Ego Development].

[a]Advanced Progressive Matrices, Strengths and Difficulties Questionnaire.

The questionnaire measures five different personality factors, which can be presumed to affect the experience of internal social problems. For the present study, two scales were included: the *Inadequacy scale* and the *Social Inadequacy scale*, which investigates self-perceived neuroticism and social behaviour. Adolescents with a high score on the *Inadequacy scale* indicate feeling generally tense, anxious (i.e. fear of failure), dismal, worried and overly sensitive. A high score on the *Social Inadequacy scale* indicates that the adolescent experiences difficulties participating in social activities and finds it difficult to make and maintain social contact (Luteijn et al., 2005). All items are graded on a four-point scale: YES = 2; ? = 1; NO = 0.

Sixthly, the Dutch version of *Loevinger's Washington University Sentence Completion Test of Ego Development* (*WUSCT*) (Westenberg et al., 2000) was completed: the *Zinnenaanvullijst Curium* (*ZALC*). The *ZALC* can be used by respondents 8 to 25 years of age. The *ZALC* consists of sentences to be completed and is usually conducted in writing. Regardless of age, respondents can complete the sentences in structurally very different ways (Loevinger, 1976, 1998). It assesses stages of ego development. The stages have a fixed sequence, and each stage indicates how one looks at oneself in relation to the world. The stages are Symbiotic, Impulsive, Self-Protective, Conformist, Self-Aware, Conscientious, Individualistic, Autonomous and Integrated (Westenberg et al., 2005).

Procedure

The adolescents were included to participate in the study after approval by the adolescents and written authorization by a parent. The students were given verbal instructions in the classroom and asked to complete the questionnaires described below independently. Written instructions were available for the teachers in the form of a manual. Test administration took 2 h and took place in a group setting. Two psychologists were continuously available and walked around to assist with questions or comments from the teachers and the adolescents.

Data analysis

IBM SPSS Statistics version 27 was used to analyse the data (IBM Corp, 2021). First, the mean scores of the subscales were calculated. Within analyses of variance (MANOVAs) (Wilks's, Allen & Bennett, 2012) were conducted to determine the effects of the levels of cognitive intelligence on the dependent variables of interest in the current study (i.e. the affective elements). The equality of variance was tested using Levene's Test (Allen and Bennett 2012). To guard against Type I errors (i.e. finding false positives due to multiple testing; Grimm & Yarnold, 2000), the alpha (α) for establishing statistical significance was set at 0.01. Given the large sample and the number of dependent variables, we avoided reporting chance findings and we kept the alpha (α) at 0.01.

Due to unequal sample sizes, post-hoc Bonferroni multiple comparison tests were applied.

RESULTS

Differences among adolescents in emotional and social well-being

First, adolescents of different intelligence levels were compared. The four groups were compared regarding self-perceived strengths and difficulties, emotional intelligence, feelings of competence, feelings of (social) (in)adequacy and ego development.

A one-way MANOVA was conducted to examine group differences in strengths and difficulties. No significant multivariate differences were found within the gifted population nor between the four groups of adolescents. The assumption of homogeneity of variance was met for all dependent variables, as indicated by Levene's tests (ps = 0.686). The intellectually highly gifted, intellectually gifted, near gifted and high-average to superior intelligent adolescents did not differ significantly in their strengths and difficulties Wilks' $\Lambda = 0.928$, $F(15,1599) = 0.72$, $p = 0.767$, $\eta_p^2 = 0.006$.

The multivariate effect for emotional intelligence (interpersonal intelligence, adaptability and positive impression) was significant, Wilks' $\Lambda = 0.947$, $F(9,1319) = 3.30$, $p = 0.001$, $\eta_p^2 = 0.018$. The assumption of homogeneity of variance was met for all dependent variables, as indicated by Levene's tests (ps = 0.066–0.984). No significant differences emerged within the gifted population nor between the groups in the post-hoc Bonferroni multiple comparison tests.

No significant multivariate differences were found for self-reported feelings of social competence, either within the gifted population or between the four groups, Wilks' $\Lambda = 0.956$, $F(12,1432) = 2.06$, $p = 0.017$, $\eta_p^2 = 0.015$.

Similarly, no significant multivariate differences were found for feelings of (in)adequacy, either within the gifted population or between the four groups Wilks' $\Lambda = 0.990$, $F(6,494) = 0.394$, $p = 0.883$, $\eta_p^2 = 0.005$. Likewise, no significant differences were observed in ego development across the cognitive capacity groups, Wilks' $\Lambda = 0.979$, $F(6,782) = 1.401$, $p = 0.211$, $\eta_p^2 = 0.011$. Because the MANOVA results were non-significant, no follow-up univariate analyses were conducted.

DISCUSSION

The aim of the present study was to obtain deeper insight into potential differences among secondary school students, situated at *the upper end of the cognitive ability* *distribution* and *attending the highest educational tracks*, including distinctions *within the gifted population*, most notably between gifted and highly gifted adolescents (in line with the harmony or disharmony hypotheses). More specifically, we wanted to gain insight into adolescents' self-perceived differences regarding their strengths and difficulties, emotional intelligence, feelings of competence, feelings of social (in)adequacy, social (in) adequacy and ego development. We expected in general no significant differences concerning the emotional and social well-being of the intellectually (highly) gifted adolescents compared to the other adolescents, with the exception of some higher scores (emotional intelligence, specifically adaptability and ego development) and one lower score, the experience of social competence of the highly gifted adolescents. Although few differences were found, no significant higher or lower scores were found. While it was expected that the intellectually highly gifted adolescents would show relatively high levels of adaptability due to their need to compensate for the almost inevitable mismatch with the environment (Robinson, 2002), no significant difference was found. Highly comparable social functioning was found within the gifted population and between all groups, varying from high-average to superior intelligent to intellectually highly gifted adolescents. The findings of the present study do not fully support the harmony hypothesis (Baudson & Preckel, 2013). The results showed that the four groups educated at the highest levels of secondary education, including gifted and highly gifted adolescents, were similarly equipped socially and emotionally, but not to an extent that exceeded that of the other groups. The findings in this study were expected to reject the disharmony hypothesis. Overall, the findings in this study showed a negative stereotype of the gifted adolescents (Bergold et al., 2021) as generally being socially vulnerable or incompetent, to be incorrect. These findings are in line with a substantial body of research including, but not limited to Terman (1925–1959), The Study of Mathematically Precocious Youth (McCabe et al., 2020), the Marburg giftedness project (Rost, 2000, 2009), the systematic review of Francis et al. (2016) and, more recently, with the research results of Paniagua-Infantes et al. (2022) and Lavrijsen and Verschueren (2023): A high IQ might even be a resilience factor for mental health problems.

Concluding, hardly any significant differences were found between year-level peers within the cognitive spectrum from the high-average and superior ranges into the highly gifted range in their emotional and social well-being. This could have to do with environmental conditions experienced by the adolescents in this study, because, like everybody else, they develop while interacting with their environment (Rinn, 2024). The equivalence in emotional and social well-being of the participating adolescents regardless of their cognitive intelligence in the present study can be explained by the reasonably similar environments in which the participating adolescents in the schools are educated and grow up: all participating adolescents attended schools and received education on the higher levels.

Strengths

The present study has several strengths, specifically that *all* eighth graders from six schools could participate in the study, preventing sampling bias. They were not selected from gifted-only programs or based on teacher recommendations (Golle et al., 2023). Intellectually highly gifted and gifted adolescents were compared with their year-level peers within the gifted population and within an academically selective environment with adolescents with high-average to superior intelligence, which provided large enough comparison groups. Using comparison groups drawn from an environment that presents similar life challenges ensures that observed differences are not driven by contextual disparities. Contextual differences may include educational level, learning experience, linguistic input, social context, expectations, school culture and the degree of academic challenge. The comparison within the gifted population makes this study relatively unique.

Limitations

This study also has some weaknesses. All participants came from the same geographic area. Because of this choice, the internal consistency is greater; therefore, the cultural diversity is less. Additionally, drop-outs or students who were not able to fully attend school did not participate in the study. This might have resulted in a sample bias, including 'healthy' adolescents, all attending school and receiving education on the highest level. For future research, it would be good to include more varied populations with, for example, greater cultural backgrounds and SES diversity.

An additional weakness is that our highly gifted group had a minimal sample size, particularly for the NPVJ scores and for females, limiting our statistical analyses due to the possibility of false positives and false negatives. Of course, in the general population the number of highly gifted adolescents is also quite limited. These adolescents can therefore be relatively isolated. In this study, this isolation may have been reduced because their enrolment in higher-level secondary education afforded them access to peers who matched them both intellectually and socially. For future research it would be recommended to include more highly gifted participants, especially females.

Covariate factors such as gender and social background may have influenced the outcomes. In this study, participants came from similar educational contexts and age ranges, which reduces large background differences between groups. Including covariates under such conditions—particularly with the small sample sizes in the highly gifted group—may reduce statistical power and yield unstable parameter estimates. Also, all groups were recruited in the same way. For this reason, we considered the groups sufficiently comparable for this exploratory group comparison. We suggest that future studies include such variables explicitly, for example using MANCOVA or regression analyses.

The selection of the four groups is based on Ruf's model, which is not as well empirically supported as other models. To date, no alternative model provides an equivalent means of differentiating degrees of giftedness in the manner proposed by Ruf's level-based classification system. While empirically supported frameworks such as the Cattell–Horn–Carroll (CHC) model or Heller's Munich Model offer scientifically robust approaches and are much more differentiated and profound to characterizing cognitive abilities; these models are designed to analyse and understand cognitive profiles and the process to exceptional performances rather than to delineate gradations within the highly gifted range. For this reason, Ruf's model was selected as the most suitable tool to offer guidelines for the cut-off scores for the present study.

The balance between the Type I and Type II error is also a limitation. Many expected findings were null effects, and the strict alpha (0.01) might increase the risk of missing real effects (Type II error). However, given the large sample and the number of dependent variables, we chose to be conservative and to avoid reporting chance findings.

Self-reports may be seen as a weakness of this study, because of subjectivity and the possible susceptibility to the effects of social desirability: the desire to make oneself look good and the inclination to overestimate one's competence or, alternatively, being overly self-aware and thus scoring lower than would otherwise be the case (e.g. heightened attention to one's flaws; Bailey, 2009). This study tried to correct this by measuring positive impressions. The adolescent participants did not rate themselves overly high or low for positive impression in the subscale Positive Impression of the emotional intelligence scale. For future research, a systemic approach could be considered, involving relevant stakeholders such as teachers and parents, as exemplified by the Marburg Giftedness Study—a 25-year (and still ongoing) longitudinal investigation into the psychosocial adjustment of gifted individuals (Rost, 2000).

Additionally, we would like to suggest a specific comparison within the gifted population, comparing highly gifted with gifted participants. By involving more informants, the risk of socially desirable responding by the adolescents themselves is reduced, as the self-reports can be compared and contrasted with input from other informants. This multi-informant approach enhances the validity of the data by allowing for cross-validation across perspectives.

MAIN CONCLUSION

Even when we narrow our analysis to the upper end of the cognitive ability distribution, the highest educational tracks, including distinctions within the gifted population, this study found no evidence for the negative stereotype that highly gifted or gifted adolescents would be socially vulnerable or maladjusted as a group. Intellectually highly gifted adolescents in this study appeared to be as emotionally and socially adapted as gifted adolescents, near gifted and high-average to superior intelligent adolescents in the Netherlands. This study contributes to the growing body of literature challenging a long-standing stereotype that gifted and also highly gifted adolescents are inherently socially and emotionally maladjusted. By empirically testing both the (dis)harmony hypothesis across varying levels of intellectual giftedness, the present findings offer a more nuanced perspective on the psychosocial functioning of intellectually highly gifted adolescents. The inclusion of ego development, emotional intelligence and multiple facets of social functioning broadens the theoretical understanding of giftedness beyond academic performance. These findings invite further research into the contextual and personal factors that may moderate or mediate the relationship between giftedness and socio-emotional development.

IMPLICATIONS

From a practical standpoint, the results have implications for how educators, psychologists and policymakers approach the support of gifted adolescents. The lack of evidence for widespread emotional and social problems in intellectually (highly) gifted students highlights the risk of overgeneralizing vulnerability and underlines the importance of individualized, strength-based perspectives (see also Lijbers et al. (2025), suggesting a strength-based approach for gifted students with learning problems). Especially since gifted adolescents often face societal prejudice (Bergold et al., 2021) and maybe the gifted participants in this study have also encountered it. Misunderstandings and an unpleasant way of being approached, too high expectations or envy, for example, stemming from societal prejudice, can result in significant social and emotional challenges for gifted individuals (Mönks et al., 2009). Indeed, repeated confrontation with negative stereotypes can stigmatize them and cause people to feel different, socially inept and weird, including (highly) gifted individuals (Grobman, 2009; Matheis et al., 2019). In order to protect themselves from the negative environmental influences, defence mechanisms, such as masking their giftedness, might be activated (Bergold et al., 2021; Frumau-van Pinxten et al., 2020; Gross, 2002; Rimm, 2002; Ruf, 2009; Wood & Laycraft, 2020). Alongside more strength-based perspectives of gifted and highly gifted people, the attention should shift also toward the broader ecological conditions that shape emotional and social outcomes. Even greater attention should be devoted to reducing or eliminating cognitive and social mismatches between highly gifted and gifted individuals and their environments.

RECOMMENDATIONS

Educational settings should aim to provide environments in which gifted students can express their full identity without the need for masking or conforming. This includes awareness of societal stereotypes, fostering psychological safety and offering appropriate challenges both academically and socially. Furthermore, the findings underscore the importance of avoiding pathologizing giftedness and instead recognizing the potential for gifted adolescents to thrive.

Psychoeducation may help increase awareness of possibly unconscious biases regarding the gifted and also the highly and profoundly gifted in society: 'Where there is interest and willingness, where there is acceptance and understanding, these children can thrive and soar' (Tolan, 2018a, 2018b).

Finally and most importantly, greater recognition and respect for the individuality of any gifted person is encouraged, in keeping with the legacy of Franz Mönks (Csermely, 2020).

AUTHOR CONTRIBUTIONS

Mia Frumau-van Pinxten: Conceptualization; methodology; data collection; data analysis; writing – original draft; overall project coordination. **Willy Peters:** Writing – review and editing; supervision. **Véronique De Gucht:** Methodology; writing – review and editing; formal analysis. **Lianne Hoogeveen:** Writing – review and editing; supervision; conceptualization; project administration. **Anouke W. E. A. Bakx:** Writing – review and editing; supervision; conceptualization; project administration.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest or competing interests.

DATA AVAILABILITY STATEMENT

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

ETHICS STATEMENT

The authors certify that the study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards.

CONSENT

Informed consent was obtained from all individual participants included in the study. Verbal informed consent as well as informed consent by email was obtained prior to the interview. Verbal informed consent was obtained to publish their data prior to submitting their paper to a journal. Written informed consent was obtained from the parents.

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